3.10. Ramification.

Def 3.4.9. (Sec(X) & Tan(X))

We define such varieties by sec bundle & tangent bundle.

$X \subseteq \mathbb{P}^r$ smooth irreducible of dim n.

consider incidence variety $\mathcal{I} = \{(x, L) \in \mathbb{P}^r \times G \mid x \in L\}$. $G = Grass(r+1, r-n)$

then $\mathcal{I}/G$ universal family $\mathcal{I} \hookrightarrow \mathbb{P}^r \times G$.

$X \hookrightarrow \mathbb{P}^r$ gives $TX \hookrightarrow T\mathbb{P}^r|_X$ $\quad 0 \to \mathcal{O} \to \mathcal{O}(1)^{r+1} \to T_{\mathbb{P}^r} \to 0$

then $TX(-1)$ lifts to $\widehat{TX} \subseteq \mathcal{O}_X^{r+1}$ (rank $n+1$) giving $X \xrightarrow{f} G$

$$\begin{array}{ccc} \mathcal{I} & \xrightarrow{\lambda_2} & \mathbb{P}^r \\ \downarrow{\scriptstyle \lambda_1} & & \\ X & \xrightarrow{f} & G \end{array}$$

$Tan(X) := \lambda_2(\lambda_1^{-1}(f(X)))$ closed subvariety.

Consider $Hilb_r X =: X^{[r]}$ universal family $\Sigma \subseteq X \times X^{[r]} \subseteq \mathbb{P}^r \times X^{[r]}$

fibre constant Hilbert poly $\Rightarrow$ $\lambda_1 : \Sigma \to X$ $\quad \lambda_2 : \mathbb{P}^r \times X^{[r]} \to X^{[r]}$

consider evaluation map $H^0(\mathcal{O}_{\mathbb{P}^r}(1) \boxtimes \mathcal{O}_{X^{[r]}}) \otimes \mathcal{O}_{\mathbb{P}^r \times X^{[r]}} \to \mathcal{O}_\Sigma(1)$

$\lambda_2^* \rightsquigarrow \mathcal{O}_{X^{[r]}}^{r+1} \to \mathcal{E}$ $\quad V = \mathbb{C}^{r+1}$

where $\mathcal{E} = \lambda_{2*}\mathcal{O}_\Sigma = \lambda_{1*}\mathcal{O}_\Sigma$ locally free of rank 2.

the adj map gives $X^{[r]} \to Gr(r+1, 2) = \{\text{lines in } \mathbb{P}^r\}$

then

$$\begin{array}{ccc} \mathcal{U} & \xrightarrow{\sim} & \mathbb{P}^r \\ \downarrow{\scriptstyle p_1} & & \\ X^{[r]} & \xrightarrow{f} & Gr(r+1, 2) \end{array}$$

$Sec(X) = p_2(p_1^{-1}(f(X^{[r]})))$ irreducible

Hilbert–Chow map $X^{[r]} \xrightarrow{\varphi} X^{(r)} = X \times X / S_r$ $\quad \Delta^{[r]} := \varphi^*(\Delta)$ $\quad Tan(X) = p_2(p_1^{-1}(f(\Delta^{[r]})))$ irreducible.

$\Delta = Im(X \to X \times X \to X^{(r)})$

Prop. 3.4.5. (Dimension of Sec(X) & Tan(X))

Either $\dim \mathrm{Sec}(X) = 2n+1$ $\dim \mathrm{Tan}(X) = 2n$ or $\mathrm{Sec}(X) = \mathrm{Tan}(X)$

proof of $\mathrm{Tan}(X) \neq \mathrm{Sec}(X)$ it suffices to show $\dim \mathrm{Tan}(X) = 2n$

(If $\dim \mathrm{Tan}(X) = 2n$, then $2n+1 \leq \dim \mathrm{Sec}(X) \leq \dim(p_1^{-1}(f(X^{reg}))) = \dim f(X^{reg}) + 1 \leq 2n+1$.)

Suppose $\dim \mathrm{Tan}(X) < 2n$, then $\exists$ linear space $\mathcal{L}$ of $\mathrm{codim} < 2n$ s.t.

$\mathcal{L} \cap X = \emptyset \Leftarrow \mathcal{L} \cap \mathrm{Tan}(X) = \emptyset$ but $\mathcal{L} \cap \mathrm{Sec}(X) \neq \emptyset$ ($\dim \mathrm{Sec}(X) > \dim \mathrm{Tan}(X)$)

projection with center $\mathcal{L}$ gives $X \xrightarrow{\varphi} \mathbb{P}^{r'}$, $r' = \mathrm{codim}\,\mathcal{L} - 1$, $r' < 2n$

φ is unramified but not inj on closed pts $\Rightarrow$ contradiction. □

Def. 3.4.7. (Local Degree)

① $f: X \to Y$ finite surj morphism of irr varieties of $\dim n$. Y smooth

then $(\Omega^1_{X/Y})_{\eta_x} = \Omega^1_{k(X)/k(Y)} = 0$ since $\mathbb{C}$ perfect. $V := X \setminus \mathrm{Supp}\,\Omega^1_{X/Y}$

$X \setminus V$ called ramification locus, $f(X \setminus V)$ called branching locus.

② (If Y smooth), $f^{-1}(Y \setminus \mathrm{Bran}(f)) \to Y \setminus \mathrm{Bran}(f)$ is étale, since

$\forall y \in Y \setminus \mathrm{Bran}(f)$ $x \in X_y$ $\dim T_y Y = \dim Y$ $\dim X \leq \dim T_x X \leq \dim T_y Y$

then $\dim X = \dim_x X \Rightarrow x \in X^{reg}$ then $f^{-1}(Y \setminus \mathrm{Bran}(f))$ smooth

Miracle Flatness $\Rightarrow f^{-1}(Y \setminus \mathrm{Bran}(f)) \to Y \setminus \mathrm{Bran}(f)$ flat, flat + unramified $\Rightarrow$ étale.

If Y not smooth, this may not be true. consider $| \to \alpha$.

② Analytically $\forall y$ $\exists U(y)$ s.t. $f^{-1}(U(y)) = \coprod_{f(x')=y} V(x')$ $V(x)$ disjoint.

$V(x') \to U(y)$ still finite (quasi-finite + closed)

$e_f(x) := \deg(V(x) \to U(f(x)))$ well-defined deg is defined on unbranching locus

($\mathrm{Bran}(f)$ closed proper in Y, then $U \setminus \mathrm{Bran}(f)$ dense in U, take $y_n \to y$

$f|_{V(x)}$ is covering map near $y_n \Rightarrow$ deg is constant, thus independant of U, V)

Obviously $\deg f = \sum_{f^{-1}(y)} e_f(x_i)$ $\forall y$. $e_f(x)$ is $\wedge$ upper semicontinuous. (analytically.)

(for $V(x) \xrightarrow{f} U(f(x))$ $\forall x' \in V(x)$ $\deg f = \sum_{f^{-1}(f(x'))} e_f(x_i) = e_f(x) \geq e_f(x')$)

③ Higher ramification locus $R_\ell(f) := \{x \in X \mid e_f(x) > \ell\}$. closed. $R_1(f) = \mathrm{Ram}(f)$

analytic top

Fact: R_ℓ is analytic set thus Zar closed

Theorem 3.48. (Ramification of Covering to Proj space)

X irreducible proj variety of dim n, $f: X \to \mathbb{P}^n$ branched covering of degree d, then $\exists x \in X$ s.t. $e_f(x) \geq \min\{d, n+1\}$.

In general, $\mathrm{codim}_X R_\ell(f) \leq \ell$ if $\ell \leq \min\{d-1, n\}$.

take $\ell = \min\{d-1, n\}$ $R_\ell(f) = \{x \in X \mid e_f(x) \geq \min\{d, n+1\}\}$ of $\mathrm{codim} \leq n \Rightarrow \dim \geq 0$

proof of $d=1$ trivial $\mathrm{codim}_X X = 0$. WLOG $d \geq 2$ $\ell \geq 1$.

induction, if $n=1$, $\ell=1$ Theorem $\Leftrightarrow R_1(f) \neq \emptyset$

if $R_1(f) = \emptyset$ f is étale, but $\pi_1^{\text{ét}}(\mathbb{P}^1) = 0$ Contradiction

Assume $n \geq 2$, $L \subseteq \mathbb{P}^n$ general hyperplane s.t. $f^{-1}(L) =: X'$ irreducible

$\exists V \subseteq \mathbb{P}^n$ dense open s.t. $f^{-1}(V) \to V$ étale, then take L s.t. $L \subsetneq \mathbb{P}^n \setminus V$, then $\forall y \in L \cap V$ $f^{-1}(L)_y = X_y$ contains d pts.

Claim. For general L, $e_{f'}(x) = e_f(x)$ $\forall x \in f^{-1}(L)$. $\operatorname{codim} R_\ell = \operatorname{codim} R'_\ell$

$\forall x \in f^{-1}(L)$, take $y_n \to f(x)$ (analytic) $y_n \in L \cap V$ then $e_f(x) = e_{f'}(x) = \lim \#\, f|_{V(y_n)}^{-1}(y)$

modify L s.t. L "intersect" X

since $f^{-1}(L) \to L$ finite, $\deg(f|_L) = d$. induction $\Rightarrow$ $f': X \to L = \mathbb{P}^{n-1}$ Theorem ✓

then $\operatorname{codim} R_\ell = \operatorname{codim} R'_\ell \le \ell$ $\forall \ell \le \min\{n-1, d-1\}$

it suffices to prove $\ell = n$ case i.e. $R_n \neq \emptyset$ $(n-1 < d-1)$

i.e. $\exists s^* \in X$ s.t. $e_f(s^*) \ge n+1$.

$\dim R_{n-1} = \dim X - \operatorname{codim} R_{n-1} \ge n - (n-1) = 1$

take a irr comp S of R_{n-1}, consider $F = f \times f|_S : X \overset{V}{\times} S \to \mathbb{P}^n \times \mathbb{P}^n$

X irr S irr $\Rightarrow X \times S$ irr $\dim V = n + \dim S \ge n+1$

F is finite, $\dim F(X \times S) \ge n+1 > n$ then by Fulton–Hansen connectedness

$F^{-1}(\Delta_{\mathbb{P}^n})$ is connected, now $\Delta_S \subseteq S \times S$ $\Delta_S \subsetneq F^{-1}(\Delta_{\mathbb{P}^n})$

$X \overset{V}{\times} S \to X$ $\dim X \overset{V}{\times} S - \dim S = \dim \Delta_S + \Delta_S$ irreducible

$\downarrow$ finite $\quad \downarrow$

$S \to \mathbb{P}^n$ then Δ_S is irr component of $F^{-1}(\Delta_{\mathbb{P}^n})$

If $\Delta_S = F^{-1}(\Delta_{\mathbb{P}^n})$ $f(x) = f(s) \Rightarrow x = s$ then $\forall s \in S$ $\#\, f^{-1}(f(s)) = 1$

$e_f(S) = d$ then $d \ge n+1 \Rightarrow S \subseteq R_n$ ✓

If $\Delta_S \subsetneq F^{-1}(\Delta_{\mathbb{P}^n})$ then $\exists$ component $T \neq \Delta_S$ $T \cap \Delta_S \neq \emptyset$

fix $(x_k, s_k) \in T \subseteq X \times S$ with $x_k \neq s_k$ $f(x_k) = f(s_k)$ $(x_k, s_k) \to (s^*, s^*) \in \Delta_S \cap T$

then $\lim x_k = \lim s_k = s^*$ $(T \setminus \Delta_S$ dense in $T)$

$s_k \in R_{n-1}$ $e_f(s_k) \ge n$ $e_f(x_k) \ge 1$ $\xrightarrow[\text{Fact}]{}$ $e_f(s^*) \ge e_f(s_k) + e_f(x_k) \ge n+1$.

$s^* \in R_n$ then n case is done. ∎